In [2]:
```python
# Imports
import warnings
import numpy as np
import pandas as pd
import pyfixest as pf
import seaborn as sns
import matplotlib.pyplot as plt
import pyreadstat
from statsmodels.nonparametric.smoothers_lowess import lowess
import statsmodels.formula.api as smf
import statsmodels.api as sm
from typing import List
from IPython.display import display, Latex, Markdown
pd.options.mode.chained_assignment = None
```

In [3]:
```python
data = pd.read_csv("https://raw.githubusercontent.com/akhandelwal8/globaleconomics/refs/heads/main/hwk/hwk5_gravity_GRC.csv", sep='\t')
```

# 1)

In [4]:
```python
data.groupby(['year', 'iso_o']).agg({'value': 'sum'}).reset_index()
```

Out[4]:

| | year | iso_o | value |
|---|---|---|---|
| **0** | 2019 | GRC | 37322.708 |

In [5]:
```python
data.groupby(['year', 'iso_o', 'iso_d']).agg({'value': 'sum'}).reset_index().sort_values('value', ascending=False).head(5)
```

Out[5]:

| | year | iso_o | iso_d | value |
|---|---|---|---|---|
| **109** | 2019 | GRC | ITA | 4085.217 |
| **60** | 2019 | GRC | DEU | 2518.291 |
| **223** | 2019 | GRC | TUR | 2191.733 |
| **57** | 2019 | GRC | CYP | 2126.615 |
| **23** | 2019 | GRC | BGR | 1746.437 |

In [7]:
```python
sortedData = data.sort_values(by='value',ascending=False)
totalTrade = np.sum(sortedData['value'])
top5frac = np.sum(sortedData['value'].head(5))/totalTrade*100
top5frac
```

Out[7]:
```
np.float64(33.942587981558034)
```

The top 5 countries - Italy, Germany, Turkey, Cyprus, and Bulgaria - make up 33.94% of Greece's total trade in 2019.

# 2)

In [8]:
```python
# Log trade
data['log_trade'] = np.log(data['value'] + 1)
```

```python
# Log distance
data['log_distance'] = np.log(data['dist'])
```

In [11]:
```python
# Regression
model1 = smf.ols('log_trade ~ log_distance', data=data).fit(cov_type="HC0")
print(model1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_trade   R-squared:                       0.326
Model:                            OLS   Adj. R-squared:                  0.322
Method:                 Least Squares   F-statistic:                     113.6
Date:                Sun, 15 Feb 2026   Prob (F-statistic):           2.92e-21
Time:                        13:27:05   Log-Likelihood:                -415.07
No. Observations:                 200   AIC:                             834.1
Df Residuals:                     198   BIC:                             840.7
Df Model:                           1
Covariance Type:                  HC0
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept       15.9570      1.239     12.879      0.000      13.529      18.385
log_distance    -1.5690      0.147    -10.660      0.000      -1.857      -1.281
==============================================================================
Omnibus:                        7.871   Durbin-Watson:                   1.982
Prob(Omnibus):                  0.020   Jarque-Bera (JB):                7.780
Skew:                           0.442   Prob(JB):                       0.0204
Kurtosis:                       2.608   Cond. No.                         84.5
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC0)
```

log_distance: As distance increases by 1%, the predicted average trade decreases by 1.57%

Intercept: If log distance is zero, the predicted average trade is 15.96 log

# 3)

In [13]:
```python
data['log_gdp_d'] = np.log(data['gdp_d'])
```

In [14]:
```python
# Regression
model2 = smf.ols('log_trade ~ log_distance + log_gdp_d', data=data).fit(cov_type="HC0")
print(model2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_trade   R-squared:                       0.734
Model:                            OLS   Adj. R-squared:                  0.731
Method:                 Least Squares   F-statistic:                     193.5
Date:                Sun, 15 Feb 2026   Prob (F-statistic):           9.42e-45
Time:                        13:27:22   Log-Likelihood:                -282.70
No. Observations:                 175   AIC:                             571.4
Df Residuals:                     172   BIC:                             580.9
Df Model:                           2
Covariance Type:                  HC0
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept        2.6088      1.342      1.944      0.052      -0.021       5.239
log_distance    -1.3452      0.110    -12.177      0.000      -1.562      -1.129
log_gdp_d        0.6579      0.047     13.881      0.000       0.565       0.751
==============================================================================
Omnibus:                       10.569   Durbin-Watson:                   2.083
Prob(Omnibus):                  0.005   Jarque-Bera (JB):               10.804
Skew:                           0.541   Prob(JB):                      0.00451
Kurtosis:                       3.557   Cond. No.                         277.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC0)
```

log_distance: As distance increases by 1% (holding gdp of destination country constant), the predicted average trade decreases by 1.35%

log_gdp_d: As the gdp of destination country increases by 1% (holding distance constant), the predicted average trade increases by 0.66%

# 4)

## a)

In [15]:
```
data['log_trade_ctrl_ms'] = data['log_trade'] - (model2.params['Intercept'] + model2.params['log_gdp_d'] * data['log_gdp_d'])
```

## b)

In [16]:
```
# Regression
model3 = smf.ols('log_trade_ctrl_ms ~ log_distance', data=data).fit(cov_type="HC0")
print(model3.summary())
```

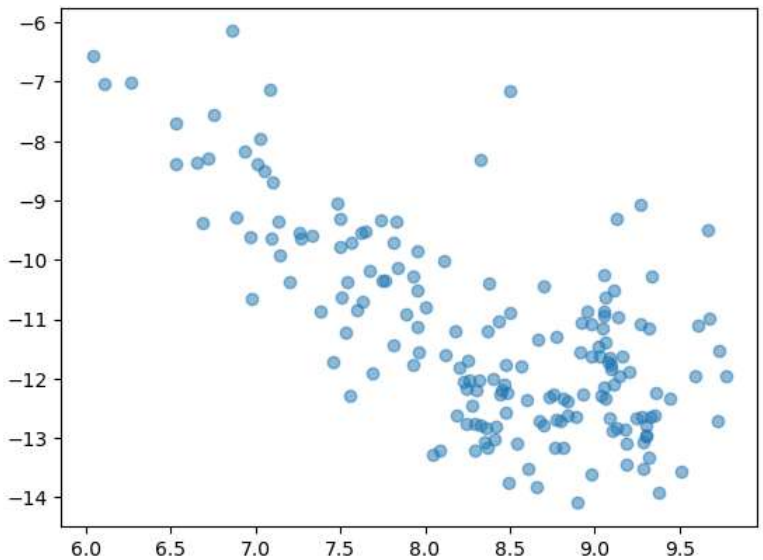
```
                            OLS Regression Results
==============================================================================
Dep. Variable:      log_trade_ctrl_ms   R-squared:                       0.467
Model:                            OLS   Adj. R-squared:                  0.464
Method:                 Least Squares   F-statistic:                     149.0
Date:                Sun, 15 Feb 2026   Prob (F-statistic):           4.03e-25
Time:                        13:27:40   Log-Likelihood:                -282.70
No. Observations:                 175   AIC:                             569.4
Df Residuals:                     173   BIC:                             575.7
Df Model:                           1
Covariance Type:                  HC0
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept     3.275e-15      0.917   3.57e-15      1.000      -1.797       1.797
log_distance    -1.3452      0.110    -12.208      0.000      -1.561      -1.129
==============================================================================
Omnibus:                       10.569   Durbin-Watson:                   2.083
Prob(Omnibus):                  0.005   Jarque-Bera (JB):               10.804
Skew:                           0.541   Prob(JB):                      0.00451
Kurtosis:                       3.557   Cond. No.                         84.5
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC0)
```

## c)

In [17]:
```
plt.scatter(data['log_distance'], data['log_trade_ctrl_ms'], alpha=0.5)
```

Out[17]:
```
<matplotlib.collections.PathCollection at 0x26e3a8112b0>
```



# 5)

In [18]:
```
data[data['comlang'] == 1]
```

Out[18]:

| | year | iso_o | iso_d | value | gdp_d | dist | border | comlang | comleg | log_trade | log_distance | log_gdp_d | log_trade_ctrl_ms |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **57** | 2019 | GRC | CYP | 2126.615 | 24900000.0 | 957.481 | 0.0 | 1.0 | 0.0 | 7.662757 | 6.864306 | 17.030378 | -6.150104 |

In [20]:
```
# Regression
model4 = smf.ols('log_trade ~ log_distance + log_gdp_d + C(border) + C(comlang)', data=data).fit(cov_type="HC0")
print(model4.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_trade   R-squared:                       0.752
Model:                            OLS   Adj. R-squared:                  0.746
Method:                 Least Squares   F-statistic:                     717.7
Date:                Sun, 15 Feb 2026   Prob (F-statistic):          1.92e-104
Time:                        13:27:58   Log-Likelihood:                -273.30
No. Observations:                 173   AIC:                             556.6
Df Residuals:                     168   BIC:                             572.4
Df Model:                           4
Covariance Type:                  HC0
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept          0.5605      1.374      0.408      0.683      -2.133       3.254
C(border)[T.1.0]   1.6649      0.354      4.708      0.000       0.972       2.358
C(comlang)[T.1.0]  3.4597      0.216     15.999      0.000       3.036       3.884
log_distance      -1.1489      0.119     -9.692      0.000      -1.381      -0.917
log_gdp_d          0.6770      0.046     14.610      0.000       0.586       0.768
==============================================================================
Omnibus:                       13.679   Durbin-Watson:                   2.038
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               15.140
Skew:                           0.604   Prob(JB):                     0.000516
Kurtosis:                       3.801   Cond. No.                         320.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC0)
```

It does not surprise me. It makes sense that countries sharing the same border and language would trade more. Also considering how significantly Greece trades with Cyprus (the only country sharing a language), the magnitude makes sense.

# 6)

In [22]:
```
EU = ['AUT','BEL','BGR','HRV','CYP','CZE','DNK','EST','FIN','FRA','DEU','GRC','HUN','IRL','ITA','LVA','LTU','LUX','MLT','NLD','POL','PRT','ROU',
      'SVK','SVN','ESP','SWE']

data['inEU'] = data['iso_d'].isin(EU).astype(int)

# Regression
model5 = smf.ols('log_trade ~ log_distance + log_gdp_d + C(border) + C(comlang) + C(inEU)', data=data).fit(cov_type="HC0")
print(model5.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_trade   R-squared:                       0.766
Model:                            OLS   Adj. R-squared:                  0.759
Method:                 Least Squares   F-statistic:                     619.1
Date:                Sun, 15 Feb 2026   Prob (F-statistic):          8.90e-106
Time:                        13:28:13   Log-Likelihood:                -268.26
No. Observations:                 173   AIC:                             548.5
Df Residuals:                     167   BIC:                             567.4
Df Model:                           5
Covariance Type:                  HC0
==================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept            -0.9716      1.519     -0.640      0.522      -3.948       2.005
C(border)[T.1.0]      2.0496      0.419      4.886      0.000       1.228       2.872
C(comlang)[T.1.0]     2.9648      0.198     14.960      0.000       2.576       3.353
C(inEU)[T.1]          0.9656      0.254      3.808      0.000       0.469       1.463
log_distance         -0.9258      0.152     -6.075      0.000      -1.224      -0.627
log_gdp_d             0.6494      0.046     14.184      0.000       0.560       0.739
==============================================================================
Omnibus:                       20.765   Durbin-Watson:                   1.988
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               25.827
Skew:                           0.775   Prob(JB):                     2.47e-06
Kurtosis:                       4.087   Cond. No.                         335.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC0)
```

By leaving the EU, Greece's exports to the EU would be cut by approximately half.